Neural Recommendation Models

CP4285 Modern Recommendation Systems — Week 04

CP4285 Instruction Team

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Scan for the slides
soc-n.us/cp4285-t2610-w04

01 From Evaluation to Model



WEEK 03 RECAP

🔑 Key Point: Evaluation Is More Than a Score

Defensible evaluation connects a protocol, a ranking metric, and the limits of the evidence.

1. Design
evaluation protocols

2. Compute
ranking metrics

3. Critique
metric selection

4. Explain
limitations of offline evaluation

☝️ **Ask what the recommender is being credited for—and what the evidence cannot establish.**

Offline ranking metrics

Metric / signal	Question answered	Useful when
Precision@K	Of the shown *K*, how many were relevant?	Visible space is scarce.
Recall@K	How much relevant content was retrieved?	Missing useful items is costly.
MAP	Are binary-relevant items repeatedly early?	Several relevant items may matter.
NDCG	Are higher-value items early?	Relevance is graded; rank matters.

🤔 **To think about**

How do precision and recall evolve as *k* increases? When is P@K or R@K more useful as an evaluation lens?

Beyond relevance: measure what receives attention

Engagement
CTR · dwell time
Did people act after a visible recommendation opportunity?

Exposure
Impression share · coverage
Who and what receives a place in the ranked list?

Discovery
Diversity · novelty · serendipity
Does the list expand useful options beyond familiar repeats?

ⓘ **Note**

These measures answer different questions. A system can earn clicks while concentrating exposure, or improve discovery while leaving CTR unchanged.

The feedback loop hidden in a log

Exposure
What the system displayed
→
Interaction
Click, play, skip, save
→
Log
Observed, selected data
→
Model
Learns from the log
→
Future exposure
Changes the next log

⚠ **Warning**

Unobserved is not random. The system's own past choices influence what can be learned and evaluated.

LAST WEEK PRE-LECTURE EXERCISE

What Does a Missing Interaction Mean?

Canvas discussion board · post before Mon, 31 Aug 23:59 SGT

Post one concise response; then reply constructively to one classmate.

> **A user has never clicked a recommended item. Give two plausible interpretations: one about preference and one about exposure or the interface.**

Neural recommenders can learn richer interaction patterns, but they still learn only from the evidence the system records.



LAST WEEK PRE-LECTURE EXERCISE · CANVAS VOICES

Three careful readings of a missing click



PRE-LECTURE SYNTHESIS

Capacity is not observability

What a non-click might mean

Disinterest, no exposure, poor placement, wrong timing, unavailable inventory, or inattention.

What today's model can change

How it combines the information that was recorded - not whether the log distinguishes those explanations.



Week 04: Neural Recommendation Models

> *A neural recommender does not replace the evaluation contract. It replaces a fixed interaction rule with a **learned function** of the representations and evidence available to it.*

Note

Today: compare latent-factor and neural interaction functions, reason about implicit-feedback training, and ask when a model's score is not yet an explanation.



Learning outcomes

Build

Describe a neural recommender as embeddings, an interaction function, and a training objective.

Compare

Contrast a latent-factor dot product with a learned non-linear interaction function.

Interpret

Identify what implicit feedback supports—and what an unobserved interaction cannot prove.

Critique

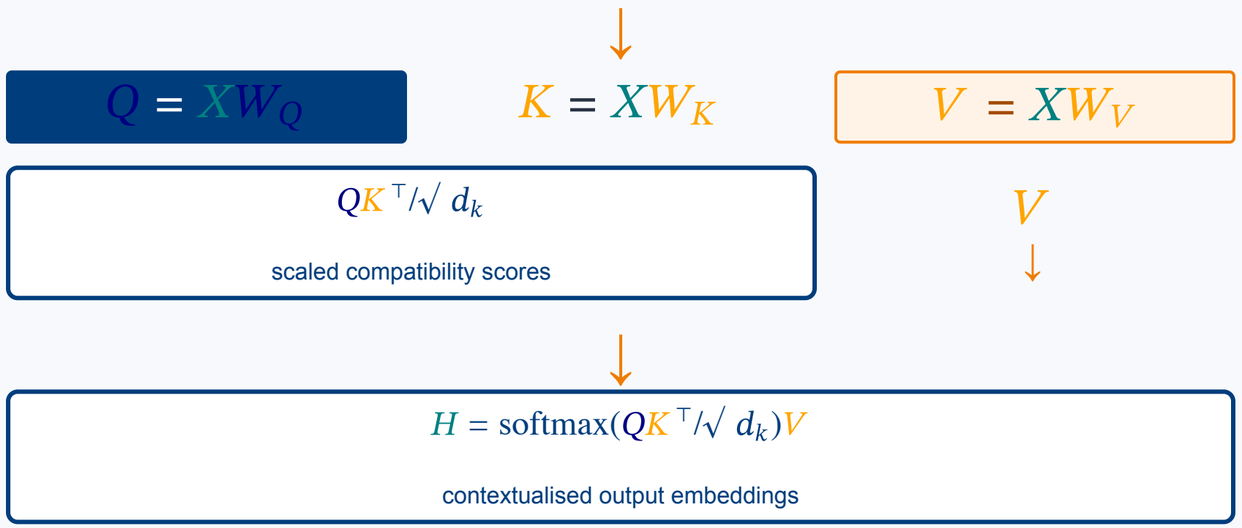
Assess whether a performance claim, explanation, and safeguard are aligned.



Transformer recap: query, key, value, attention

Scaled dot-product attention

input embeddings X



Compatibility

$$s_j = q^\top k_j$$

Queries score keys. Higher = closer match.

Weighted information

$$h = \sum_j \alpha_j v_j, \qquad \alpha = \text{softmax}(s)$$

Softmax gives weights; combine values.

🤔 **To think about**

Week 05 foreshadowing: Transformers natively deal with ordered interactions, which we don't touch this week.

Attention as a user–item matching mechanism

user embedding e_u → **user query q_u** ↔ item key/value (k_i, v_i)

Learned projections

$$q_u = W_Q e_u$$

$$k_i = W_K e_i, \quad v_i = W_V e_i$$

The user supplies the query; each candidate item supplies a key and value.

One user–item compatibility

$$q_u^\top k_i$$

A higher score means an item key better matches a user query. Across candidates, softmax turns scores into weights for values.

For this bridge, user = query and item = key/value. In a full Transformer, these are operational roles—not permanent identities.

02 From Matrix Factorisation to Neural CF



Matrix factorisation: a dot product is an interaction

Learn latent representations

$$\mathbf{p}_u \in \mathbb{R}^d, \quad \mathbf{q}_i \in \mathbb{R}^d$$

The axes are learned embeddings of one-hot users and items.

Choose a fixed interaction

$$\hat{r_{ui}} = \mathbf{p}_u{}^\top \mathbf{q}_i + b_u + b_i$$

A dot product rewards aligned representations.

Note

Use $f_\theta(p_u, q_i)$ to capture combinations a dot product misses. It needs more data, compute, and overfitting control.



Worked MF score: one user, one candidate

Given learned coordinates

$$\mathbf{p}_u = [0.8, 0.2]$$

User: high on coordinate 1; low on coordinate 2.

$$\mathbf{q}_i = [0.9, 0.1]$$

Item: also high on coordinate 1; low on coordinate 2.

These are learned coordinates, not named traits.

Multiply, then add

$$\hat{r_{ui}} = 0.8(0.9) + 0.2(0.1)$$
$$= 0.72 + 0.02 = \mathbf{0.74}$$

0.72: dominant alignment on coordinate 1.
0.02: small alignment on coordinate 2.
0.74: relative ranking score—not probability or explanation.

🔑 Key Point: NeuMF combines GMF + MLP

GMF

$$\mathbf{p}_u \odot \mathbf{q}_i$$

Weights coordinate-wise matches, extending the dot-product path.

NeuMF

1 · retain the GMF path

2 · add the MLP path

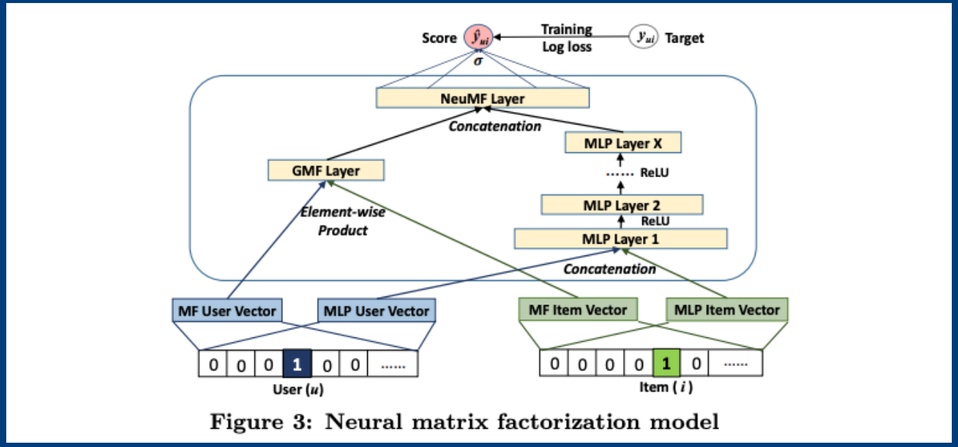


Figure 3: Neural matrix factorization model

3 · concatenate both paths.

MLP

$$f(\mathbf{p}_u, \mathbf{q}_i)$$

Mixes all coordinates through learned non-linear layers.

Worked interaction functions: same inputs, three scores

Shared toy inputs: $\mathbf{p}_u = [.8, .2]$, $\mathbf{q}_i = [.9, .1]$

1 · GMF

Element-wise match

$$[.8, .2] \odot [.9, .1] = [.72, .02]$$

Read out

$$s_G = [1, 1]^\top [.72, .02]$$

$$s_G = \mathbf{.74}$$

2 · MLP

Build a hidden representation

$$\mathbf{h} = \mathrm{ReLU}([.8 + .1, .2 + .9]) = [.9, 1.1]$$

Read out

$$s_M = [.2, .4]^\top [.9, 1.1]$$

$$s_M = \mathbf{.62}$$

3 · NeuMF

Concatenate both paths

$$\mathbf{z} = [.72, .02; .9, 1.1]$$

Read out

$$s_N = [.5, .5, .1, .1]^\top \mathbf{z} = .36 + .01 + .09 + .11$$

$$s_N = \mathbf{.57}$$

Note

Navy vectors are toy **readout weights**. Each dot product produces one scalar: s_G, s_M, or s_N.

Code lens: where each NeuMF component lives

Annotated forward pass

```
p_g, q_g = user_gmf[u], item_gmf[i]
g = p_g * q_g        # GMF path

p_m, q_m = user_mlp[u], item_mlp[i]
h = mlp(concat(p_m, q_m))  # MLP path

r_ui = readout(concat(g, h)) # scalar score
```

GMF features g + MLP features h → score r_{ui}

Two representation paths

Separate GMF/MLP embeddings create g and h on distinct paths.

One final readout

$$\hat{r_{ui}} = \mathbf{w}^\top [g; h] + b$$

$\mathbf{w}$ maps concatenated g, h to one ranking score.

03 Training for Ranked Recommendations



Implicit feedback creates a sampling decision

Observed positive

Aisha clicked **Botanic Gardens**.

A signal of possible interest—not a rating of every alternative.

Unobserved candidate

Aisha did not click **Riverside Food Court**.

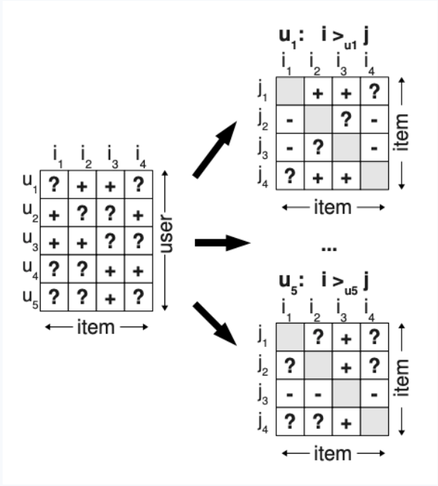
It may have been unseen, unavailable, badly timed, or unappealing.

logged click → **positive** + **sampled unobserved item** → **training comparison**

Negative sampling is an assumption about missing feedback—not a discovered dislike.



Pointwise prediction loss vs. pairwise ranking loss



Click figure to open the source paper.
Rendle et al. (2009), Fig. 1

Pointwise: predict one interaction label

$$L_{\text{pt}} = -[\, r_{ui} \log \sigma(\hat{r_{ui}}) + (1 - r_{ui}) \log(1 - \sigma(\hat{r_{ui}})) \,]$$

Training claim: “Did Aisha interact with City Gardens?”

Pairwise: predict one ordering

$$L_{\text{BPR}} = -\log \sigma(\hat{r}(u, i^{+}) - \hat{r}(u, i^{-}))$$

Training claim: “Should City Gardens rank above Riverside Food Court?”

The figure makes the choice visible: **pointwise calibrates a pair**; **pairwise orders a comparison**.

Rendle, S., Freudenthaler, C., Gantner, Z., & Schmidt-Thieme, L. (2009). *BPR: Bayesian personalized ranking from implicit feedback*. In *Proceedings of the Twenty-Fifth Conference on Uncertainty in Artificial Intelligence* (pp. 452–461). AUAI Press. PDF

BPR example: make the observed item rank higher

BPR = Bayesian Personalised Ranking. It is a personalised ranking loss for tuning model parameters so that each user's observed items score above sampled alternatives.

Triplet

u = Aisha; i^+ = Botanic Gardens (clicked); i^- = Riverside Food Court (sampled).
$s(u, i^+) = 1.20$, $s(u, i^-) = 0.40$

1 · Margin

$$\Delta = 1.20 - 0.40 = 0.80$$

Positive above sampled alternative.

2 · Pairwise probability

$$\sigma(\Delta) = \sigma(0.80) = 0.690$$

Model confidence in this ordering.

3 · Loss and update

$$L_{\text{BPR}} = -\ln(0.690) = 0.371$$

$\partial L/\partial \Delta = -0.310$: gradient descent increases the positive score and/or lowers the sampled score.

BPR ranks the click above this sampled comparison; it does not call the unclicked item "bad."

👥 Which negative is defensible?

Aisha clicked Botanic Gardens. Which sampled comparison is most defensible?

1. A restaurant that was never shown.
2. One shown after it closed.
3. One shown beside Botanic Gardens at the same time.

Groups of 3–4: choose one; give one reason and one remaining uncertainty. Prepare two sentences.



▶ Start Timer

Bandit Time: What does BPR optimise?

For Aisha's clicked Botanic Gardens and sampled Riverside Food Court, which update most directly reduces the BPR loss?

A. Widen the margin: raise Botanic Gardens' score and/or lower Riverside Food Court's score for Aisha.

B. Raise both item scores by the same amount because both may be relevant.

C. Optimise Botanic Gardens' standalone click probability without using the sampled alternative.

D. Set Riverside Food Court to a zero target for everyone who has not clicked it.

Break — 5 minutes

WHILE YOU PAUSE

☕ TriRank

TriRank extends the user–item graph with review aspects, modelling users, items, and the aspects mentioned in reviews as a tripartite graph. That extra evidence can support explanations such as "recommended because you value battery life and display quality."

TriRank was one of the final precursor works to Neural Collaborative Filtering (NeuCF) pioneered here at NUS.

He, Chen, Kan & Chen (2015), "TriRank: Review-aware Explainable Recommendation by Modeling Aspects," CIKM 2015

05:00

▶ Start Timer

04 Practical Comparison: Baseline First



The experiment contract

Hold fixed	Why	Inspect
Data + split	Both models face the same evidence.	Is test later than training?
Negative sampler	Sampling changes the ranking task.	Which unseen items become negatives?
Metrics + K	Metrics answer stated list-quality questions.	Does K match the product surface?
Budget + reporting	A gain needs cost and reproducibility context.	Time, parameters, seeds, examples.

> ⚠ **Warning**
>
> Changing the model and the split, sampler, or metric together does not identify what produced an apparent improvement.



Practical walkthrough: BPR-MF in RecBole

w04-ranking.yaml

```yaml
# reproducible random choices
seed: 2026
# ranking evaluation protocol
eval_args:
  group_by: user
  order: TO
  split: {RS: [0.8, 0.1, 0.1]}
  mode: full
# one unobserved comparison per positive
train_neg_sample_args:
  distribution: uniform
  sample_num: 1
# report top-k list quality
metrics: [Recall, NDCG]
topk: [10]
```

Illustrative run

BPR learns a pairwise objective with latent-factor, dot-product scoring.

```
run_recbole(model="BPR", dataset="ml-100k", config_file_list=["w04-ranking.yaml"])
```

Keep fixed: split · sampler · metrics · seed policy → **Change:** model = BPR → **Record:** NDCG@10 · Recall@10 · time · parameters



Practical walkthrough: change only the interaction function

S26 reference · BPR-MF

```
run_recbole(model="BPR", dataset="ml-100k", config_file_list=
["w04-ranking.yaml"])
```

Same runner from Slide 26, kept deliberately muted.

S27 implementation · NeuMF

```
run_recbole(model="NeuMF", dataset="ml-100k",
config_file_list=["w04-ranking.yaml"])
```

Only the interaction model changes.

What changes inside the scorer: NeuMF combines GMF and MLP paths. More parameters do not make implicit feedback less ambiguous.

Note

Locate the embedding tables, MLP layers, and prediction layer. Which is absent from a dot-product baseline?
🤞 Hint: See the pictorial of NeuMF on Slide 14.



👥 Deployment decision clinic

Should this service replace BPR-MF with NeuMF?

Quality signal: NDCG@10 rises from **0.214** (BPR-MF) to **0.221** (NeuMF).

Operational cost: runtime rises from **2.1** to **7.8 minutes**; parameters from **1.2M** to **5.9M**.

Group outcome: submit **one decision**—deploy NeuMF, retain BPR-MF, or collect more evidence—and justify it with **one reason that weighs both conditions**.



▶ Start Timer

Bandit Time: Which evidence supports NeuMF?

Which evidence best supports replacing BPR-MF with NeuMF in this service?

A. NeuMF wins on one seed, although the latency measurement uses a different candidate set.

B. A post-hoc explanation is persuasive to reviewers, but quality and operational cost are not compared.

C. NeuMF gains after the time split and negative sampler both change from the BPR-MF run.

D. Under the same split, sampler, and seeds, NeuMF has a stable meaningful gain, fits the latency budget, and documents limits.

Debrief: read the result as a trade-off

Quality

Stable across seeds, periods, and segments?

Operations

Fits latency, memory, and maintenance budget?

Evidence

Does the protocol justify the user-facing claim?



05 Revisiting our Card Lenses



♠ Spades: Requirements need policy 📶

stated requirement → eligibility + constraint policy → **BPR-MF or NeuMF ranking** → faithful explanation

BPR-MF

It learns an observed-versus-sampled ordering. It does not guarantee that an important attribute or rule is satisfied.

NeuMF

A flexible interaction score neither makes an attribute a hard condition nor proves that its rationale is truthful.

Overclaim

“This item is recommended because the model understands your preferences.”

Accountable claim

“Among items meeting the stated requirements, this item ranked highly under our declared data, objective, and policy.”

⚠ Warning

Use the model to rank eligible options; do not use its score as evidence that a critical requirement has been met.



♥ Hearts: Neither score identifies free preference

social cue or peer activity → prominent exposure → logged click

BPR-MF · pairwise claim

The clicked item moves above a sampled alternative. The loss does not identify independent interest, a friend's cue, or ranking-induced visibility.

NeuMF · richer patterns

It may learn complex user, item, and context co-occurrences. Flexibility cannot turn observational social traces into causal preference evidence.

Evidence + protection

Record exposure and relevant social context; compare contexts; give users controls over social cues and sharing.

What in the data lets us say this reflects preference rather than system or peer influence?

⚠ Warning

A high score may predict behaviour after influence—not explain what the person independently wants.

♦ Diamonds: Better ranking can reduce discovery

sparse long-tail evidence → BPR-MF or NeuMF ranking → top-K exposure

BPR-MF · tail evidence

Tail items have few observed positives. Sampled unobserved items become training alternatives despite weak evidence that they were seen.

NeuMF · no discovery objective

Richer signals may help where they exist, but a small aggregate NDCG gain can still concentrate exposure on familiar items.

Report beside NDCG@K

Catalogue coverage, long-tail share, new-item exposure, and relevant segment outcomes—plus candidate policy and sampler.

Would this result still count as an improvement if the top-K list became less discoverable?

⚠ Warning

BPR and NeuMF optimise the ranking objective supplied to them; neither automatically protects long-tail opportunity.

♣ Clubs: BPR and NeuMF can both learn an attack



A pairwise win is not a robustness guarantee.

BPR-MF · loss sees

Observed positive **>** sampled unobserved item. Coordinated fake positives can create favourable training pairs for a target item.

NeuMF · capacity changes

It can fit richer interaction patterns—including an attack pattern if that pattern appears in the log. More parameters are **not** an abuse defence.

Before a deployment claim

Stress-test plausible shill or coordinated behaviour. Report ranking shifts, affected users, and target-item exposure.

If suspicious accounts or concentrated bursts are removed, does the claimed gain still hold?

> ⚠ **Warning**
>
> Neither BPR-MF nor NeuMF is robust merely because it ranks held-out interactions well.



Summary

🔑 Architecture and objective are separate choices.

1 · Represent
Embeddings provide inputs.

2 · Interact
MF dot product or neural function.

3 · Optimise
BPR trains a relative ranking claim.

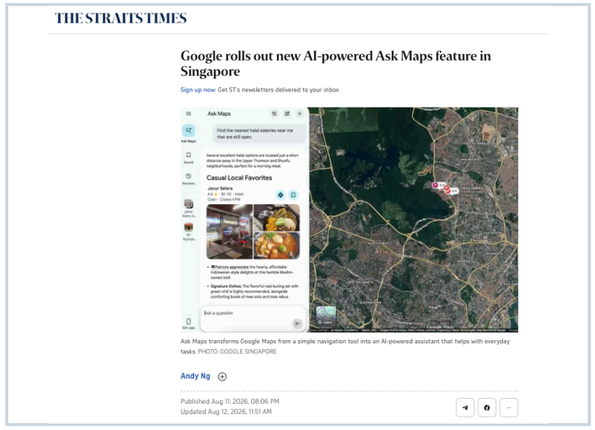
4 · Justify
Keep costs, safeguards, and limits visible.

Next week: order, recency, and session context enter the model.



NEXT WEEK PRE-LECTURE EXERCISE

Where Does One Session End?



THE STRAITS TIMES

Google rolls out new AI-powered Ask Maps feature in Singapore

Sign up now: Get ST's newsletters delivered to your inbox

Ask Maps

Find the nearest halal eateries near me that are still open.

Several excellent halal options are located just a short distance away in the Upper Thomson and Bhurtu neighborhoods, perfect for a morning meal.

Casual Local Favorites

Jalur Sutera

Ask a question

Ask Maps transforms Google Maps from a simple navigation tool into an AI-powered assistant that helps with everyday tasks. PHOTO: GOOGLE SINGAPORE

Andy Ng

Published Aug 11, 2026, 08:06 PM
Updated Aug 12, 2026, 11:51 AM

Optional reading: *The Straits Times*, 11 Aug 2026

Canvas discussion board · post before Mon, 7 Sep 23:59 SGT

Post one **100–140 word** decision memo. No classmate reply is required.

> **A person asks Ask Maps for a 2pm lunch route, explores nearby cultural sites, then later asks for a halal eatery open after 10pm. Does the late-evening query begin a new session? Answer yes or no and cite two trace details.**

Next week: sequences make order, recency, and local context part of the recommendation signal.

NEXT WEEK PREVIEW — WEEK 05

Sequential and Session-Based Recommendation

A static user profile treats interactions as a collection. A sequence model asks whether **order, recency, and session context** change the next useful recommendation.

In Week 5:

- model temporal preferences;
- predict the next item; and
- critically assess engagement-driven objectives.



Photo: Markus Spiske / Unsplash